

PES UNIVERSITY — DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
Software Engineering Lab (CS351) | Lab 3: Architectural Justification Document

NAME: Akash M P

SRN: PES1UG24CS040

Problem Statement: #40 – Student Startup Crowdfunding Platform (Retail, E-Commerce & Finance)

1. Architectural Selection

“We chose the Layered (3-Tier) Architecture for the Student Startup Crowdfunding Platform.”

2. Architectural Pattern Analysis & Comparison

Architecture	Structure & Organization	Key Advantages	Suitability for Problem #40
Layered (3-Tier) [Selected]	Organized into Presentation, Business Logic, and Data layers.	Simple to understand, clear separation of tasks, easy to manage and test.	Best Fit: Keeps milestone verification and fund release rules organized in one place.
Microservices	Multiple independent small services communicating over the network.	Can scale individual services separately, isolated failures.	Too Complex: Handling milestone approvals and payouts across multiple separate services adds unnecessary network trouble.
Client-Server	Two-tier setup where the client directly connects to the database or server.	Very simple to set up with minimal components.	Unsuitable: Lacks a proper business logic layer to protect escrow funds and handle payment security.

3. Two Specific Scenario-Related Reasons

- **Handling Milestone Verification and Controlled Fund Release:** In our student startup platform, founders do not receive all funding immediately. Pledges are held safely in escrow and released in stages (tranches) only after the student uploads milestone proof and the Faculty Committee formally reviews and approves it (FR-001, FR-005). The Layered Architecture places this verification logic inside the Business Logic Layer (Milestone Verification and Pledge & Fund Management components). This ensures that fund disbursement rules are enforced on the server and founders cannot bypass faculty approval through the user interface.
- **Separating Different User Responsibilities:** Our system serves three different user groups with distinct responsibilities: Student Founders (create campaigns, submit milestone proofs), Campaign Backers (browse projects, select reward tiers, make pledges), and Faculty Committee (review milestone evidence and approve tranche releases). The Layered Architecture keeps the user interface separate from the backend business components and database, making the platform easy to build, test, and maintain without one feature breaking another.

4. Security Advantage (Payment Protection & Role-Based Access)

- **Payment Isolation and Role Checking:** By using a dedicated Payment Service Component, card details are handled through secure tokenization directly with the external Payment Gateway (NFR-001) and are never stored in our platform database. In addition, the Business Logic Layer checks user roles before processing requests, preventing regular backers or student founders from accessing faculty approval screens or modifying fund records.

5. Performance Benefit (Fast Local Execution & Campaign Caching)

- **Direct Component Communication & Read Caching:** Because all components run within a single organized 3-tier application, milestone approvals trigger fund release steps directly through simple method calls without the network delays of microservices. Furthermore, public campaign browsing data (FR-003) can be cached in the application layer, giving students and backers fast loading times during campus funding events.